EXPERIMNET 2

1. **AIM:**
   To perform ETL and data transformation using Microsoft Power BI.

2. **OBJECTIVES:**
   - To understand the basic stages of ETL in Power BI.
   - To perform data cleaning and transformation using Power Query.
   - To combine and organize data from different worksheets.
   - To establish relationships between tables in the data model.
   - To create calculated measures using DAX.
   - To develop a basic analytical report from the transformed data.

3. **TOOLS / SOFTWARE REQUIRED:**
   - Microsoft Power BI Desktop
   - Global Superstore Dataset.xlsx
   - Power Query Editor – built into Power BI Desktop
   - DAX (Data Analysis Expressions)

4. **DATASET USED:** Global Superstore Dataset.xlsx.

5. **THEORY:**

   ETL (Extract, Transform, Load) is a data preparation process used to collect data from a source, modify it into a suitable format, and load it into an environment for analysis.

   Extract refers to obtaining data from a source such as an Excel workbook. In this experiment, the Orders, Returns, and People worksheets are imported into Power BI.

   Transform involves preparing the extracted data for analysis. Common transformations include removing duplicate or blank records, changing data types, renaming columns, creating new columns, and standardizing values. Power Query provides tools for performing these transformations and records them as applied steps.

   Load is the process of applying the transformations and loading the prepared tables into the Power BI data model. The model can then contain relationships between tables for analysis. Power BI provides modeling features for creating and managing table relationships.

DAX (Data Analysis Expressions) is a formula language used in Power BI for creating calculations such as measures and calculated columns. Measures are dynamic calculations whose results can change according to the filters and selections in a report.

In this experiment, ETL is used to prepare the Global Superstore data, while DAX measures are used to calculate important business metrics such as sales, profit, profit margin, and order count.

- <u>**PROCEDURE:**</u>

Part A: Extract – Connect to Dataset

Step 1: Download the Global Superstore dataset.

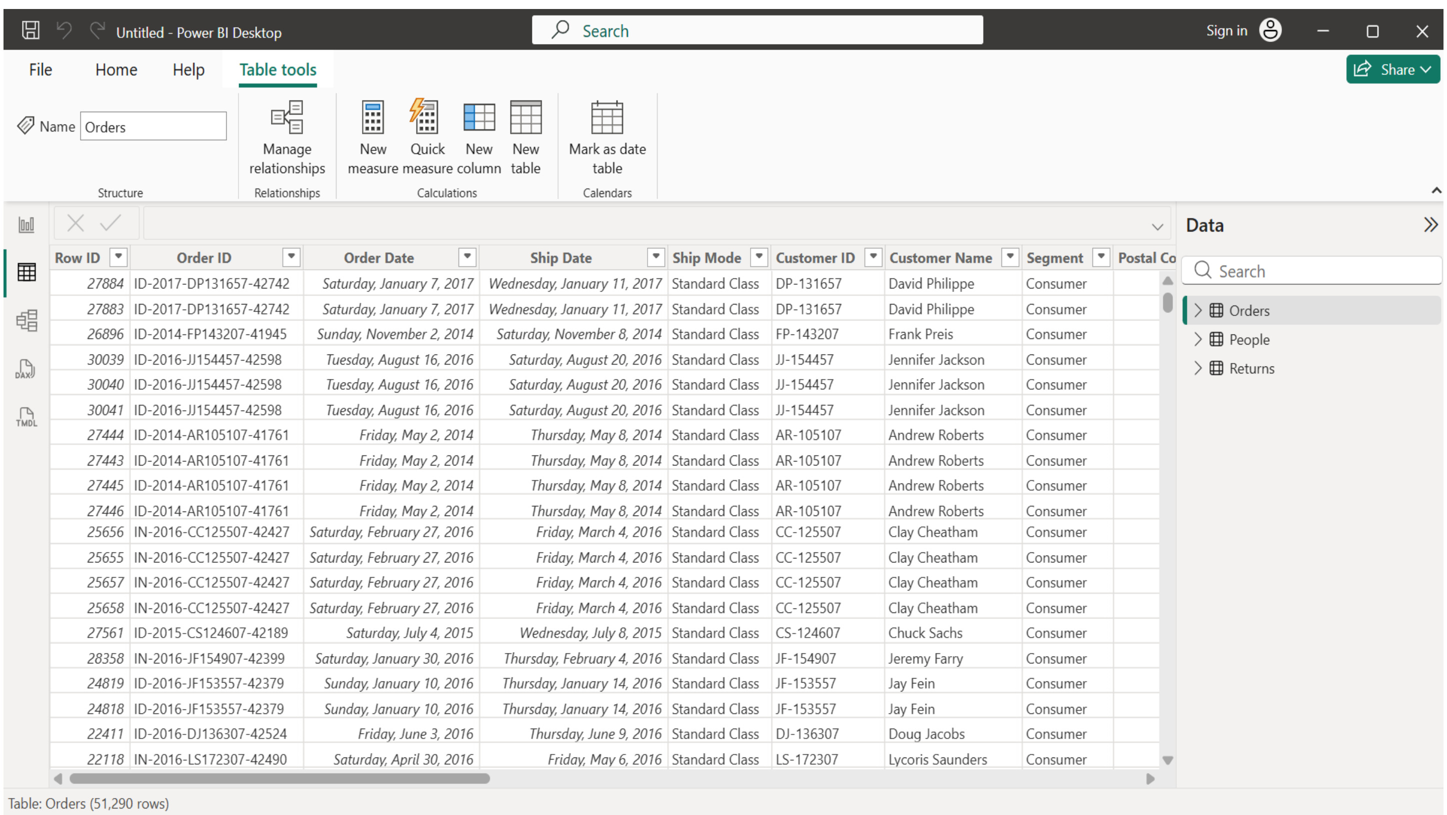
Download the Global Superstore Dataset.xlsx file and save it in the experiment folder.

Step 2: Open Power BI Desktop and connect to the Excel dataset.

Home > Get Data > Excel Workbook

Select Global Superstore Dataset.xlsx > Transform Data

Step 3: Import the required worksheets.



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Postal Co
27884	ID-2017-DP131657-42742	Saturday, January 7, 2017	Wednesday, January 11, 2017	Standard Class	DP-131657	David Philippe	Consumer	
27883	ID-2017-DP131657-42742	Saturday, January 7, 2017	Wednesday, January 11, 2017	Standard Class	DP-131657	David Philippe	Consumer	
26896	ID-2014-FP143207-41945	Sunday, November 2, 2014	Saturday, November 8, 2014	Standard Class	FP-143207	Frank Preis	Consumer	
30039	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
30040	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
30041	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
27444	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27443	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27445	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27446	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
25656	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25655	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25657	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25658	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
27561	ID-2015-CS124607-42189	Saturday, July 4, 2015	Wednesday, July 8, 2015	Standard Class	CS-124607	Chuck Sachs	Consumer	
28358	IN-2016-JF154907-42399	Saturday, January 30, 2016	Thursday, February 4, 2016	Standard Class	JF-154907	Jeremy Farry	Consumer	
24819	ID-2016-JF153557-42379	Sunday, January 10, 2016	Thursday, January 14, 2016	Standard Class	JF-153557	Jay Fein	Consumer	
24818	ID-2016-JF153557-42379	Sunday, January 10, 2016	Thursday, January 14, 2016	Standard Class	JF-153557	Jay Fein	Consumer	
22411	ID-2016-DJ136307-42524	Friday, June 3, 2016	Thursday, June 9, 2016	Standard Class	DJ-136307	Doug Jacobs	Consumer	
22118	IN-2016-LS172307-42490	Saturday, April 30, 2016	Friday, May 6, 2016	Standard Class	LS-172307	Lycoris Saunders	Consumer	

Select the following worksheets:

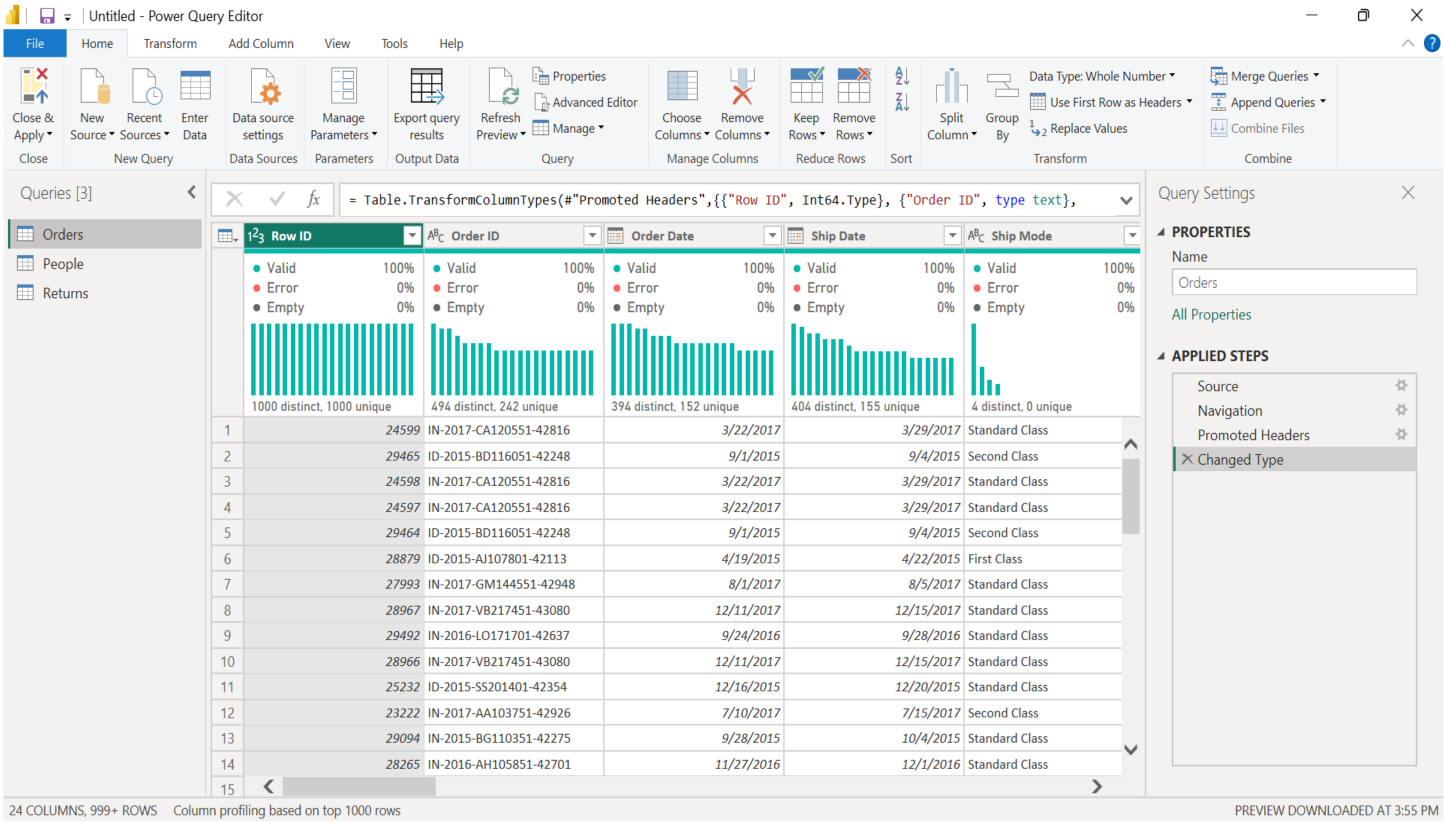
- Orders
- Returns
- People

Verify that all three tables appear in the Queries pane of Power Query Editor.

Part B: Transform – Power Query Editor

Step 4: Open Power Query Editor.

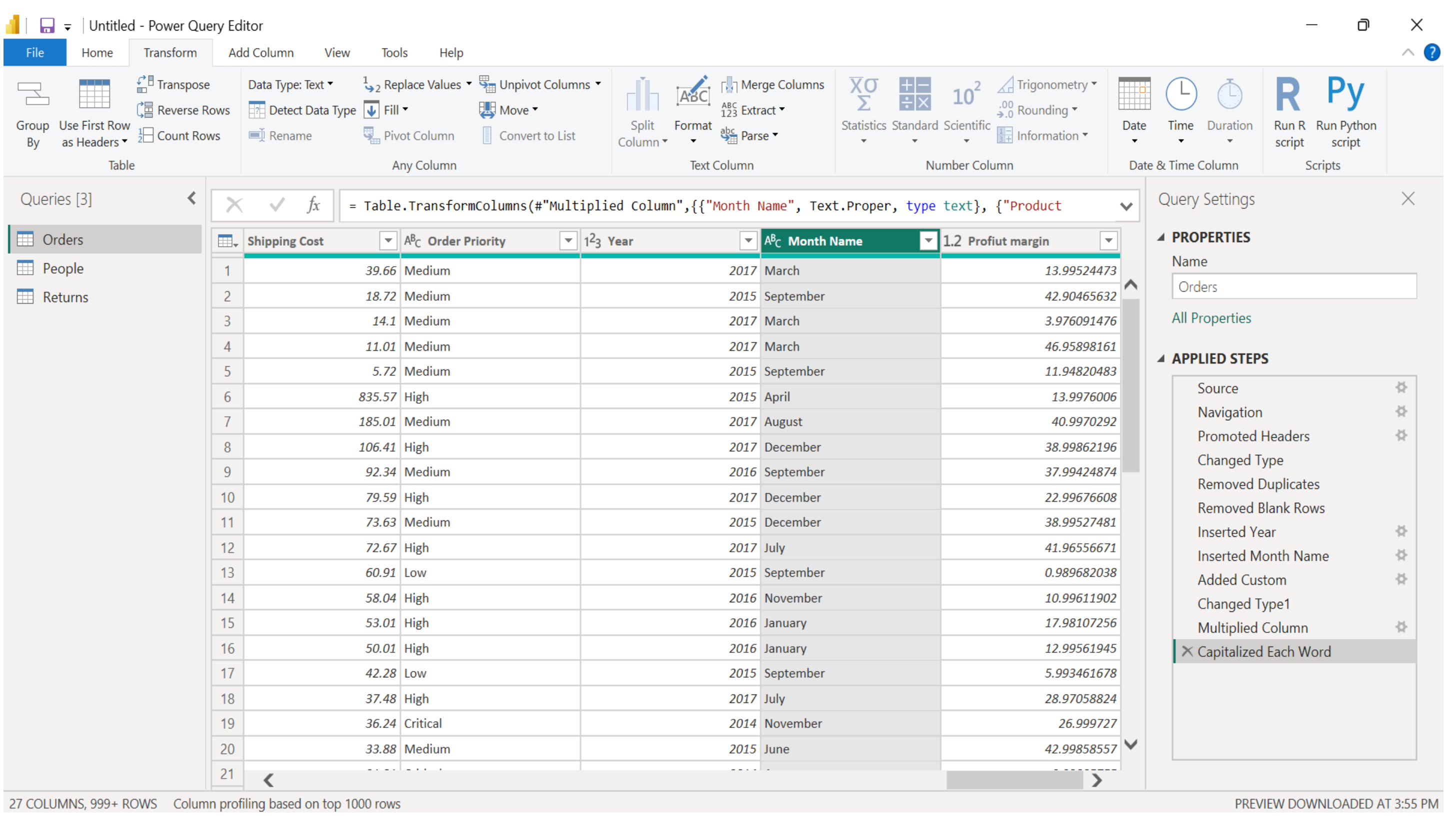
Home > Transform Data



Select the Orders table.

Step 5: Transform the Orders table.

Apply the following transformations:



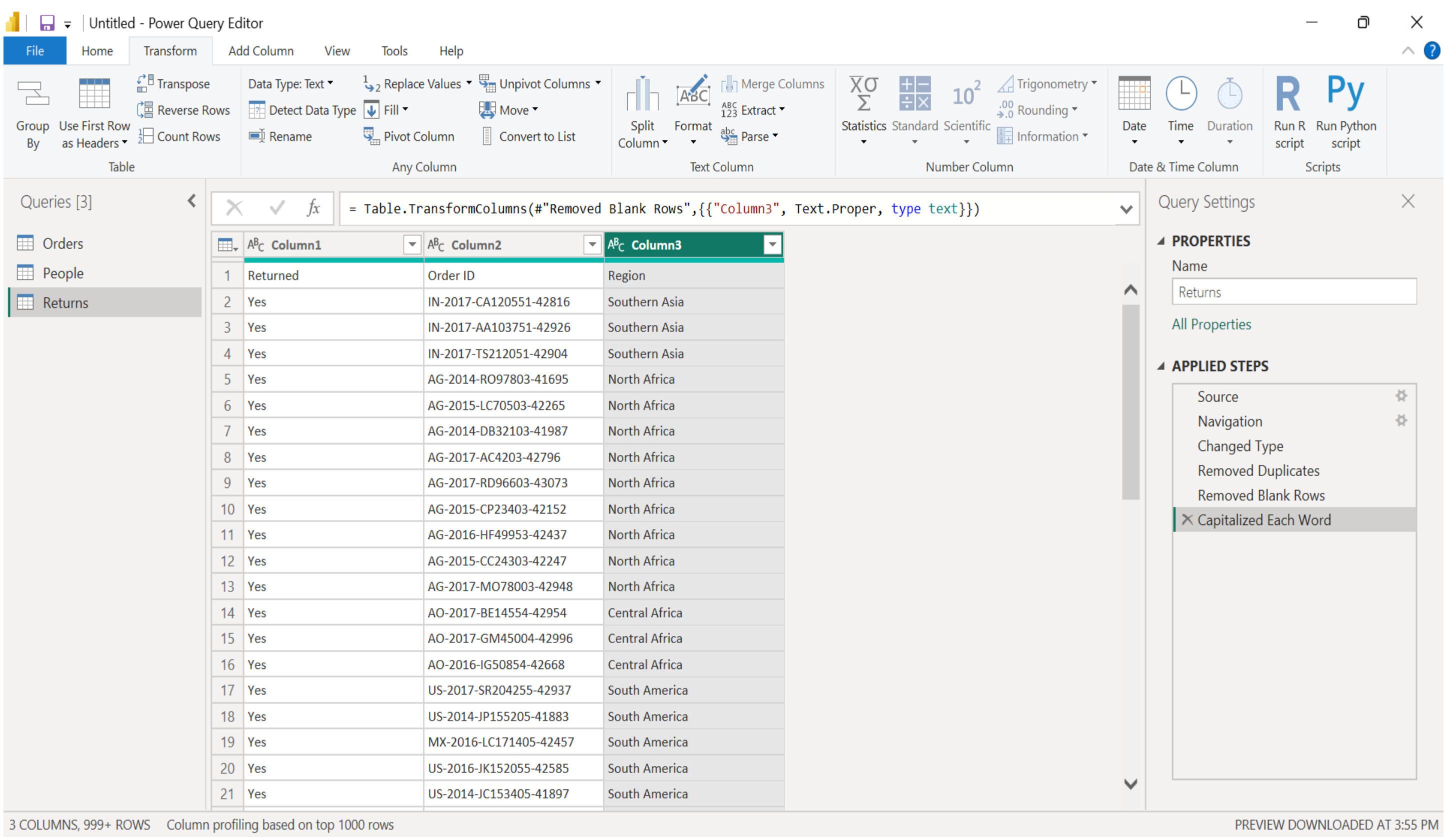
- Remove duplicate records.
- Remove blank rows.
- Verify and correct data types.
- Rename columns where required.
- Extract Year and Month from Order Date.
- Create a Profit Margin derived column.
- Standardize Region and Segment values.

Review the transformations in the Applied Steps section.

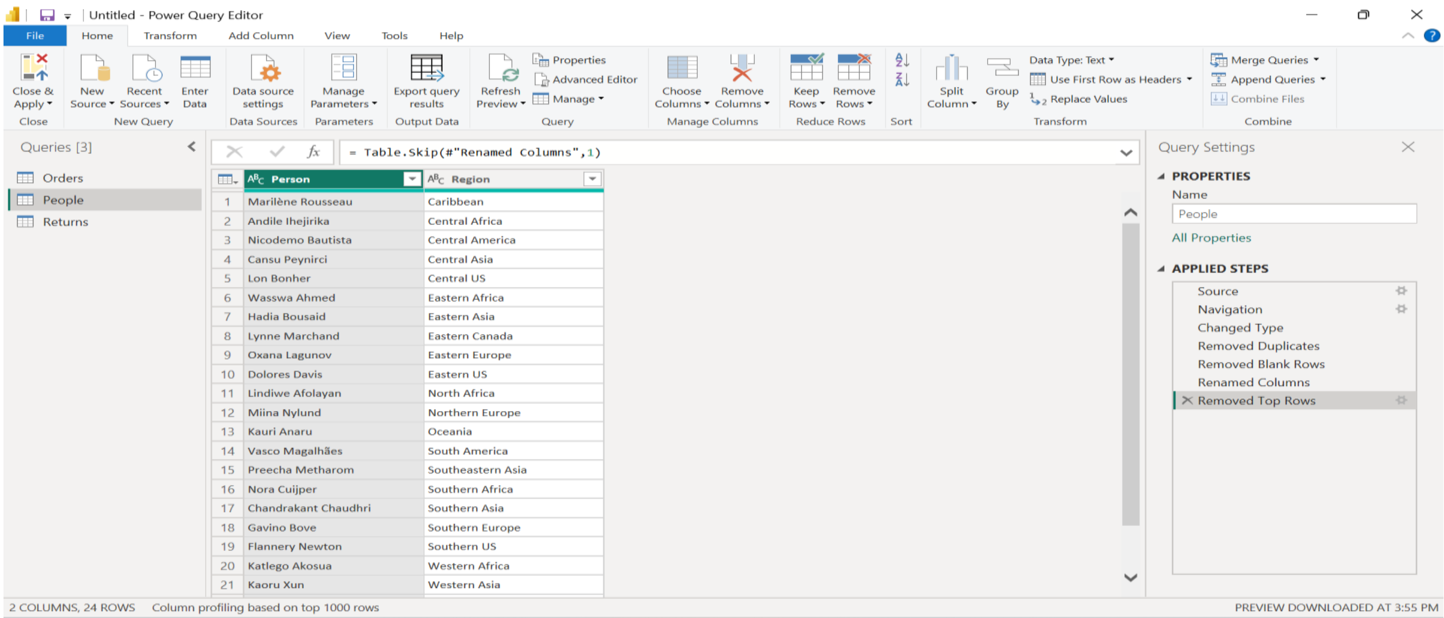
Step 6: Transform the Returns table.

Select the Returns table and:

- Remove duplicate Order IDs.
- Check for missing values.
- Standardize the Returned field values.



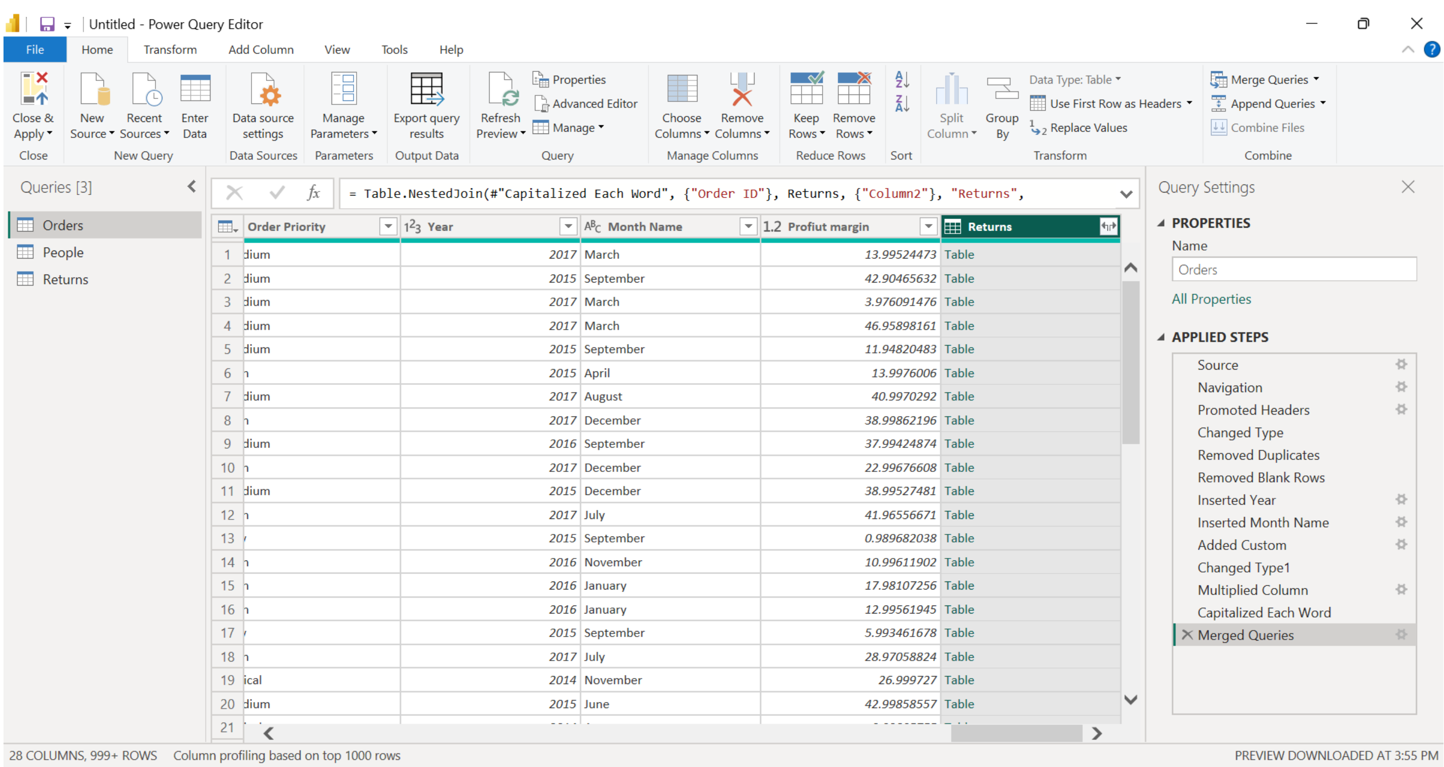
Step 7: Transform the People table.



Select the People table and:

- Validate regional assignments.
- Remove unnecessary columns.
- Verify consistency of region names.

Step 8: Merge the Orders and Returns tables.



Use the Order ID field to merge the two tables.

Orders > Merge Queries > Returns > Order ID

Select the appropriate join condition and apply the merge.

Step 9: Review and apply the transformations.

Review all Applied Steps for the three tables.

Click:

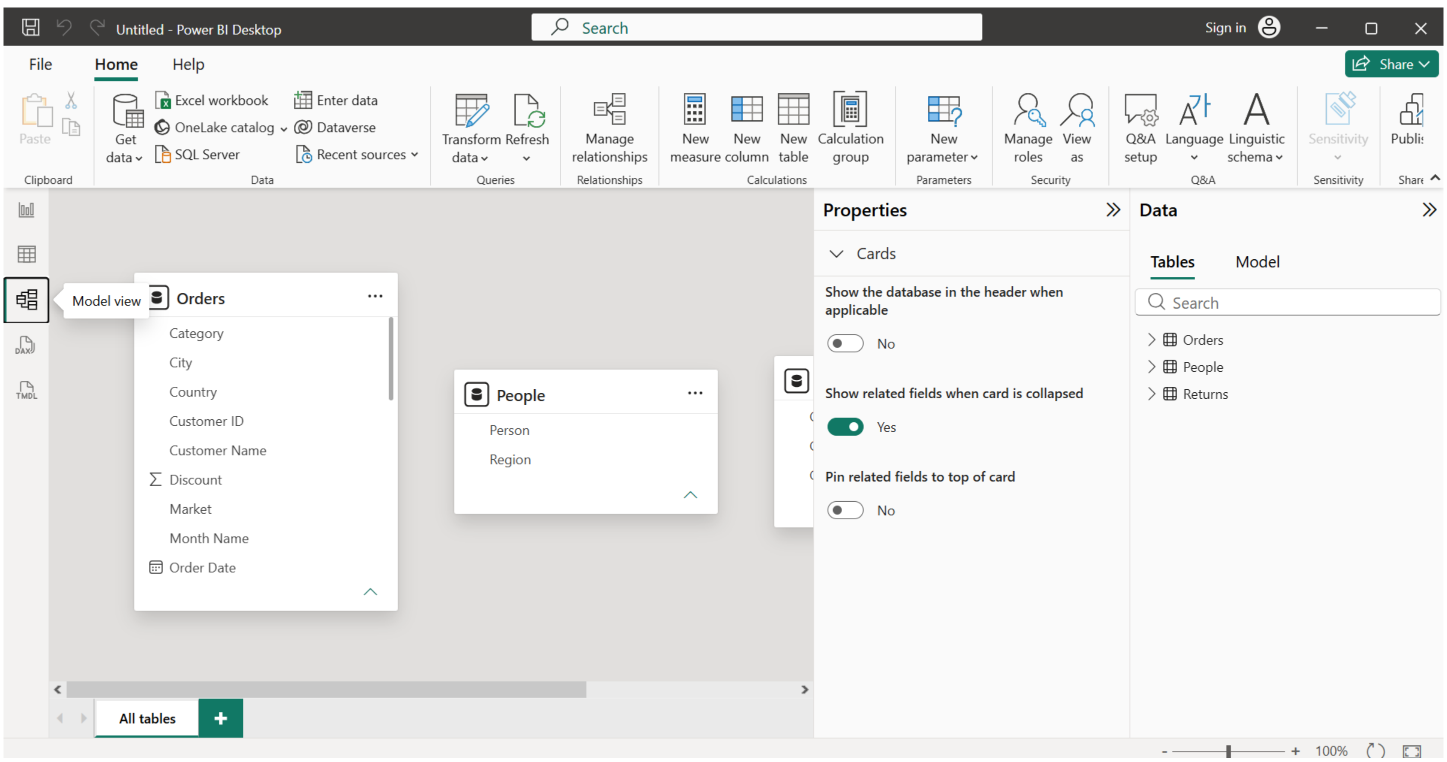
Home > Close & Apply

The transformed data is then loaded into the Power BI data model.

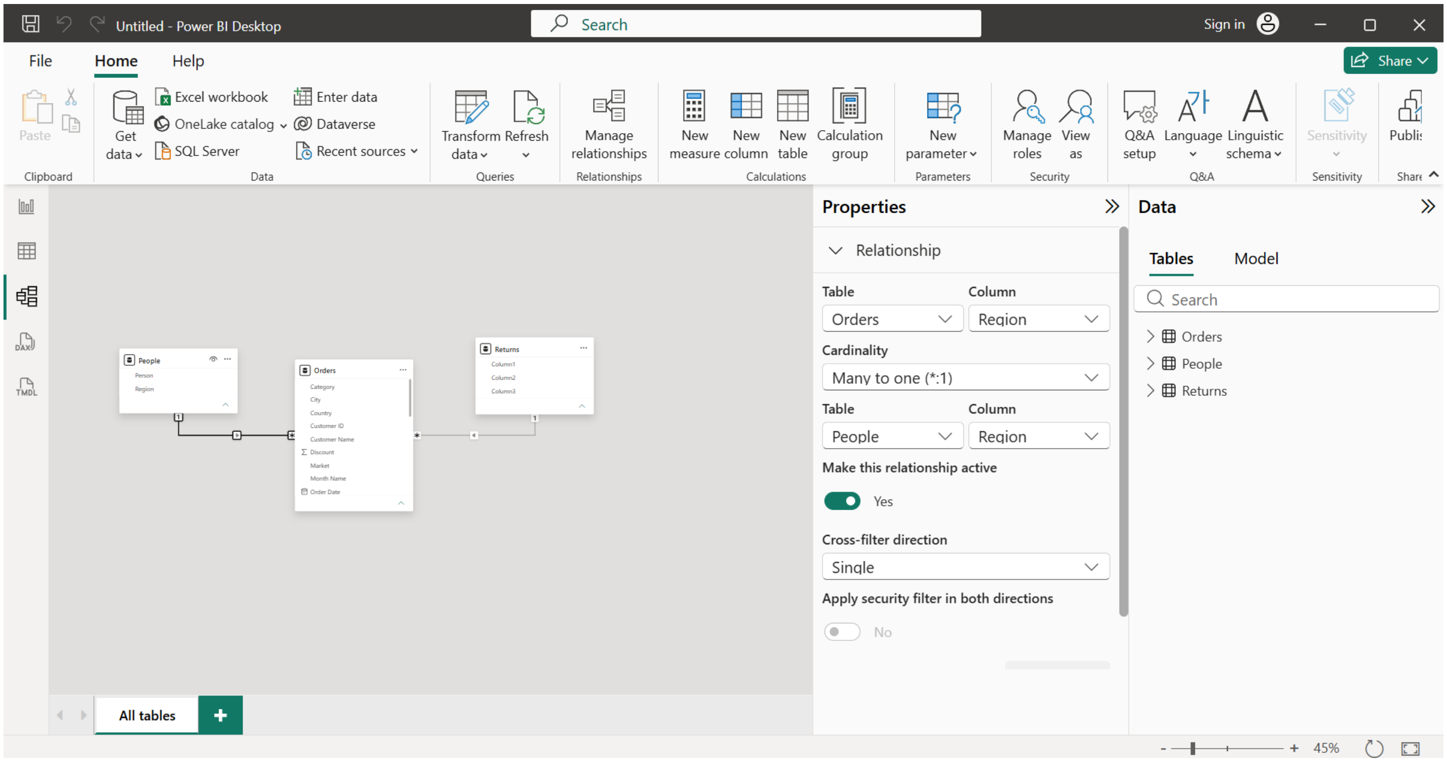
Part C: Load – Build Data Model

Step 10: Open Model View.

Select the Model View icon from the left navigation pane in Power BI.



Step 11: Create relationships between the tables.



Create the following relationships:

- Orders ↔ Returns using Order ID
- Orders ↔ People using Region

Verify that the relationships are correctly established in the model.

Step 12: Create DAX measures.

Select the Orders table and create the following measures using New Measure:

1. Total Sales = SUM(Orders[Sales])
2. Total Profit = SUM(Orders[Profit])
3. Profit Margin = DIVIDE(SUM(Orders[Profit]),SUM(Orders[Sales]))
4. Total Orders = DISTINCTCOUNT(Orders[Order ID])

These measures are used to perform analytical calculations and display key business performance indicators.

Step 13: Create the BI report/dashboard.

Create a Power BI report using the transformed data and DAX measures.

The final dashboard includes:

- Total Sales
- Total Profit
- Profit Margin
- Total Orders
- Sales by Category bar chart

- Profit by Category pie chart
- Profit Margin vs Discount scatter chart
- Detailed Sales & Profit Analysis table

Add the dashboard title:
GLOBAL SUPERSTORE – ETL & SALES ANALYSIS

6. RESULT

The Global Superstore data was successfully extracted and transformed using Power Query. The required tables were cleaned, combined, and loaded into the Power BI data model. Relationships and DAX measures were created successfully, and the transformed data was used to prepare an analytical Power BI report.

7. LEARNING OUTCOMES

- Learned the practical workflow of ETL in Power BI.
- Gained experience in transforming raw business data using Power Query.
- Learned how to prepare multiple tables for analysis.
- Understood how relationships connect tables within a Power BI data model.
- Gained practical knowledge of creating DAX measures.
- Learned how dynamic measures can be used in Power BI reports.
- Learned how transformed data can be used to create business-oriented visualizations.
- Developed practical skills in preparing an interactive analytical report.
- Understood how data preparation and modeling improve the quality of business analysis.